

Chapter 16.5: Parametric Surfaces

G5: Parameterization. I can find parametric equations for common curves, such as line segments, graphs of functions of one variable, circles, and ellipses. I can match given parametric equations to Cartesian equations and graphs. I can parameterize common surfaces, such as planes, quadric surfaces, and functions of two variables.

V4: Surface Integrals. I can set up and compute surface integrals for scalar and vector valued functions.

Mechanics

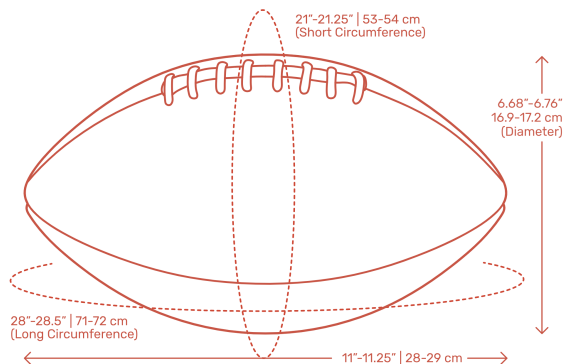
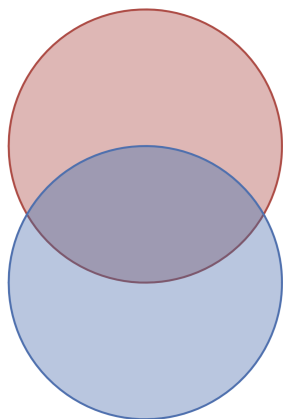
1. Consider the surface cut from the parabolic cylinder $y = 4 - x^2$ by the planes $z = 0$, $z = 2$, and $y = 0$. Sketch S and find a parameterization of S .
2. Consider the surface given by $x^2 + y^2 + z^2 = 4$, where $z \geq \sqrt{3}$. Sketch S , parametrize S and find the surface area of S .
3. Find the area of the surface cut from the “nose” of the paraboloid $x = 1 - y^2 - z^2$ by the yz -plane.
4. Find the area of the part of the surface $z = xy$ that lies within the cylinder $x^2 + y^2 = 1$.

Applications

5. Recall from Calculus class that one can form a surface by revolving the graph of a *non-negative* function $y = f(x)$ about the x -axis. Translating this into the language of parameterized surfaces, convince yourself that a parameterization can be given by

$$r(x, \theta) = (x, f(x) \cos \theta, f(x) \sin \theta)$$

[Hint: Each x -cross section is a circle with what radius?] A typical (American) football can be roughly modeled as the revolution of the area of intersection of the two circles $x^2 + (y - 1)^2 = 4$ and $x^2 + (y + 1)^2 = 4$ about the x -axis. Set up a surface integral to compute the surface area of a football. [Warning: this is not to scale!!]



Extensions

6. The tangent plane at a point $P_0 = (f(u_0, v_0), g(u_0, v_0), h(u_0, v_0))$ on a parameterized surface $\mathbf{r}(u, v) = \langle f(u, v), g(u, v), h(u, v) \rangle$ is the plane through P_0 with normal vector equal to $\mathbf{r}_u(u_0, v_0) \times \mathbf{r}_v(u_0, v_0)$.

Use this to find an equation to the tangent plane of the surface parameterized by $\mathbf{r}(r, \theta) = (r \cos(\theta))\mathbf{i} + (r \sin(\theta))\mathbf{j} + r\mathbf{k}$, with $r \geq 0, 0 \leq \theta \leq 2\pi$ at the point where $(r, \theta) = (2, \pi/4)$.

What is a Cartesian equation for this surface? Sketch it and the tangent plane.

7. The notion of curvature for curves that we explored in Chapter 13.4 has an extension for parametric surfaces, and is fundamental in a field of math called *differential geometry*. Recall from the above problem that the *unit* normal vector to a parameterized surface $\mathbf{r}(u, v)$ is given by the vector $\mathbf{n} = \frac{\mathbf{r}_u \times \mathbf{r}_v}{\|\mathbf{r}_u \times \mathbf{r}_v\|}$. Consider the following scalar quantities for the surface $\mathbf{r}(u, v)$:

$$E = \mathbf{r}_u \cdot \mathbf{r}_u$$

$$F = \mathbf{r}_u \cdot \mathbf{r}_v$$

$$G = \mathbf{r}_v \cdot \mathbf{r}_v$$

$$L = \mathbf{r}_{uu} \cdot \mathbf{n}$$

$$M = \mathbf{r}_{uv} \cdot \mathbf{n}$$

$$N = \mathbf{r}_{vv} \cdot \mathbf{n}$$

The *Gaussian curvature* of the surface is given by the quantity

$$\kappa = \frac{LN - M^2}{EG - F^2}$$

Verify that the Gaussian curvature at any point on a sphere of radius r is $\frac{1}{r^2}$. Does this result remind you of how the curvature of certain curves are also constant?